

Design and Model Summary Report

Design Overview

DESIGN Full Factorial	RUNS 8	REPLICATES 0
CENTER POINTS 0	CONTINUOUS 3	

Process Factors

Factor	Type	Settings	Coded levels
X1	Continuous	9 to 11	-1, 1
X2	Continuous	3.6 to 4.4	-1, 1
X3	Continuous	9 to 11	-1, 1

Design matrix

The complete design matrix in actual factor units is provided in Appendix A.

Model Structure

Responses: viscosity

Included Terms

Intercept	Yes
Linear Terms	X1, X2, X3
2-Term Int.	X1 : X2, X1 : X3, X2 : X3
Quadratic Terms	Not included
3-Term Int.	X1 : X2 : X3

Model Matrix Diagnostics

Metric	Value
Observations	8
Columns	8
Rank	8
Aliased columns	0
Intercept	Yes
Regression DF	7
Residual DF	0
Status	Full rank, Saturated

Saturated model

The model has no residual degrees of freedom, so residual variance and related tests are not interpretable.

General Diagnostics

Variance Inflation Factors (VIF)

VIF < 5 is low, 5-10 is elevated, and VIF > 10 indicates high multicollinearity.

Term	VIF	Flag
X1	1	Low
X2	1	Low
X3	1	Low
X1:X2	1	Low
X1:X3	1	Low
X2:X3	1	Low
X1:X2:X3	1	Low

Leverage

Average leverage: 1

Run	Leverage
1	1
2	1
3	1
4	1
5	1
6	1
7	1
8	1

Response Conditions

Optimization limits and goals configured for each response.

Response	Lower Limit	Upper Limit	Goal
viscosity	46	No	Maximize

Response Analysis: viscosity

Model Metrics

R^2	Adj. R^2	Q^2 (LOO)	PRESS (LOO)	RMSE	RMSE (LOO)
1	N/A	N/A	N/A	1.26e-14	N/A

Coefficient Summary

Term	Coefficient	Std. Error	95% CI half-width	Lower CI	Upper CI	p-value
Int	49.45	N/A	N/A	N/A	N/A	N/A
X1	-1.8	N/A	N/A	N/A	N/A	N/A
X2	-0.6	N/A	N/A	N/A	N/A	N/A
X3	0.25	N/A	N/A	N/A	N/A	N/A
X1:X2	-0.85	N/A	N/A	N/A	N/A	N/A
X1:X3	0.4	N/A	N/A	N/A	N/A	N/A
X2:X3	1.9	N/A	N/A	N/A	N/A	N/A
X1:X2:X3	1.25	N/A	N/A	N/A	N/A	N/A

Model F-test

F-value	F-crit (95%)	F-crit (99%)	p-value
N/A	N/A	N/A	N/A

Not interpretable

The F-test is undefined, commonly because model or residual degrees of freedom are insufficient.

Lack-of-Fit F-test

Not applicable

No replicated-run lack-of-fit test is applicable.

ANOVA

SD = sqrt(MS). Only the Residuals row reports residual standard deviation; RMSE = sqrt(SS_res / n).

Source	SS	dof	MS	SD
Total	77.74	7	11.1057	3.3325
Regression	77.74	7	11.1057	3.3325
Residuals	1.26e-27	0	N/A	N/A

Replicate Summary

Not applicable

No replicated runs are present.

Observed and Predicted Responses

Run	Observed	Predicted
1	51.8	51.8
2	50.2	50.2
3	51	51

Run	Observed	Predicted
4	52	52
5	51.6	51.6
6	46.6	46.6
7	42.4	42.4
8	50	50

Appendix A - Design Matrix

Run	X1	X2	X3
1	9	3.6	9
2	9	3.6	11
3	9	4.4	9
4	9	4.4	11
5	11	3.6	9
6	11	3.6	11
7	11	4.4	9
8	11	4.4	11